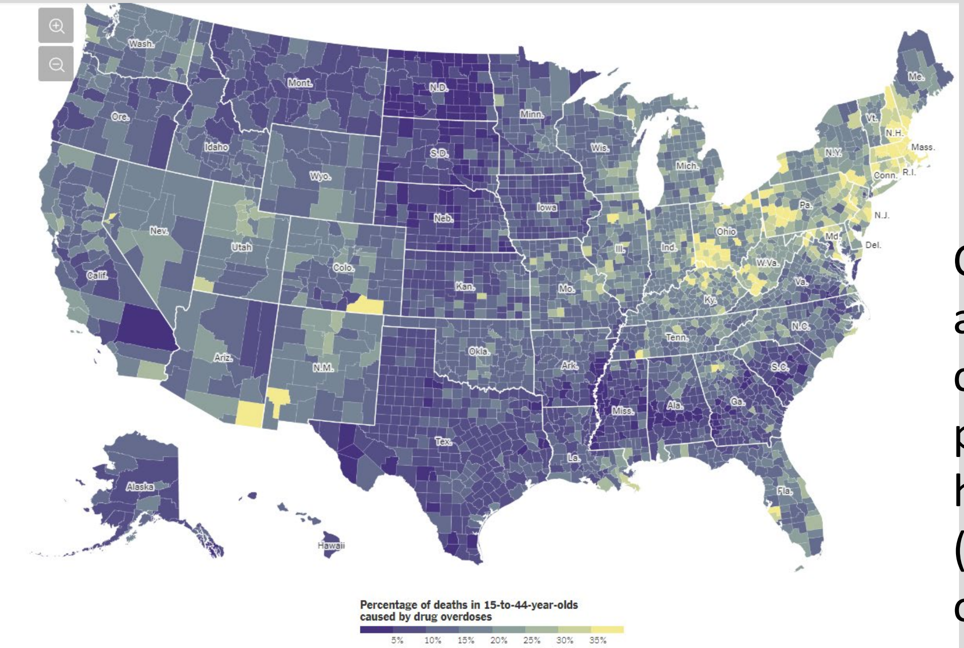
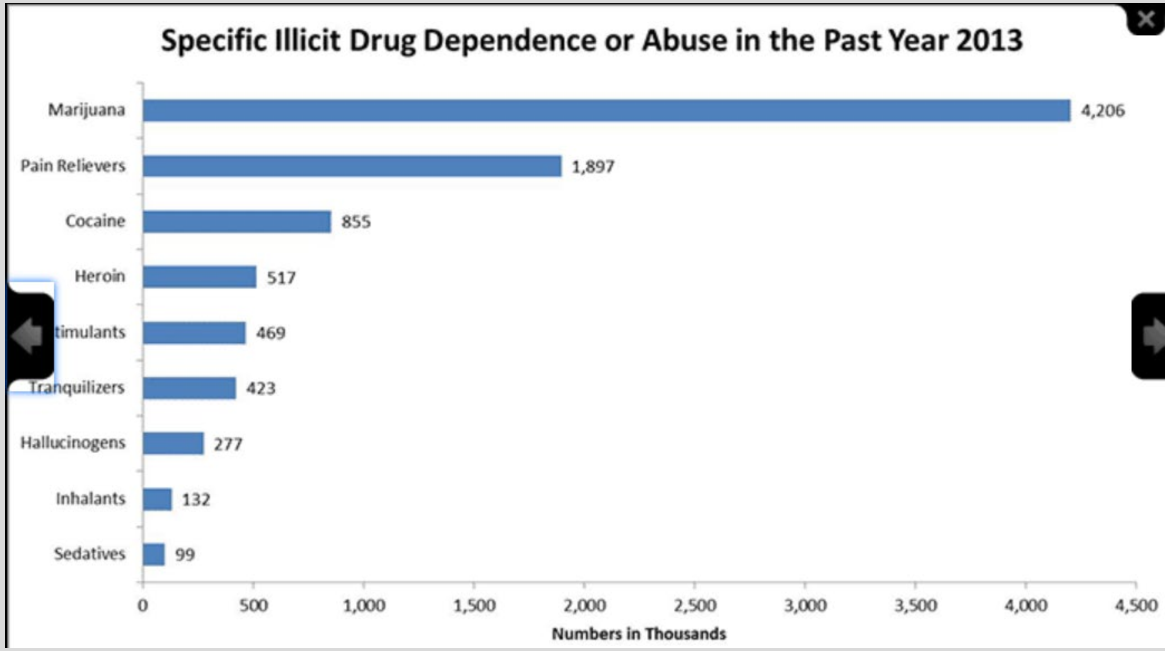
Epidemiology of Drug Addiction

- In 2016, roughly 64,000 people died from drug overdoses; one of leading causes of **accidental death**
- Most people use drugs for the first time when they are **teenagers**
- More than half of new illicit drug users begin with **marijuana**



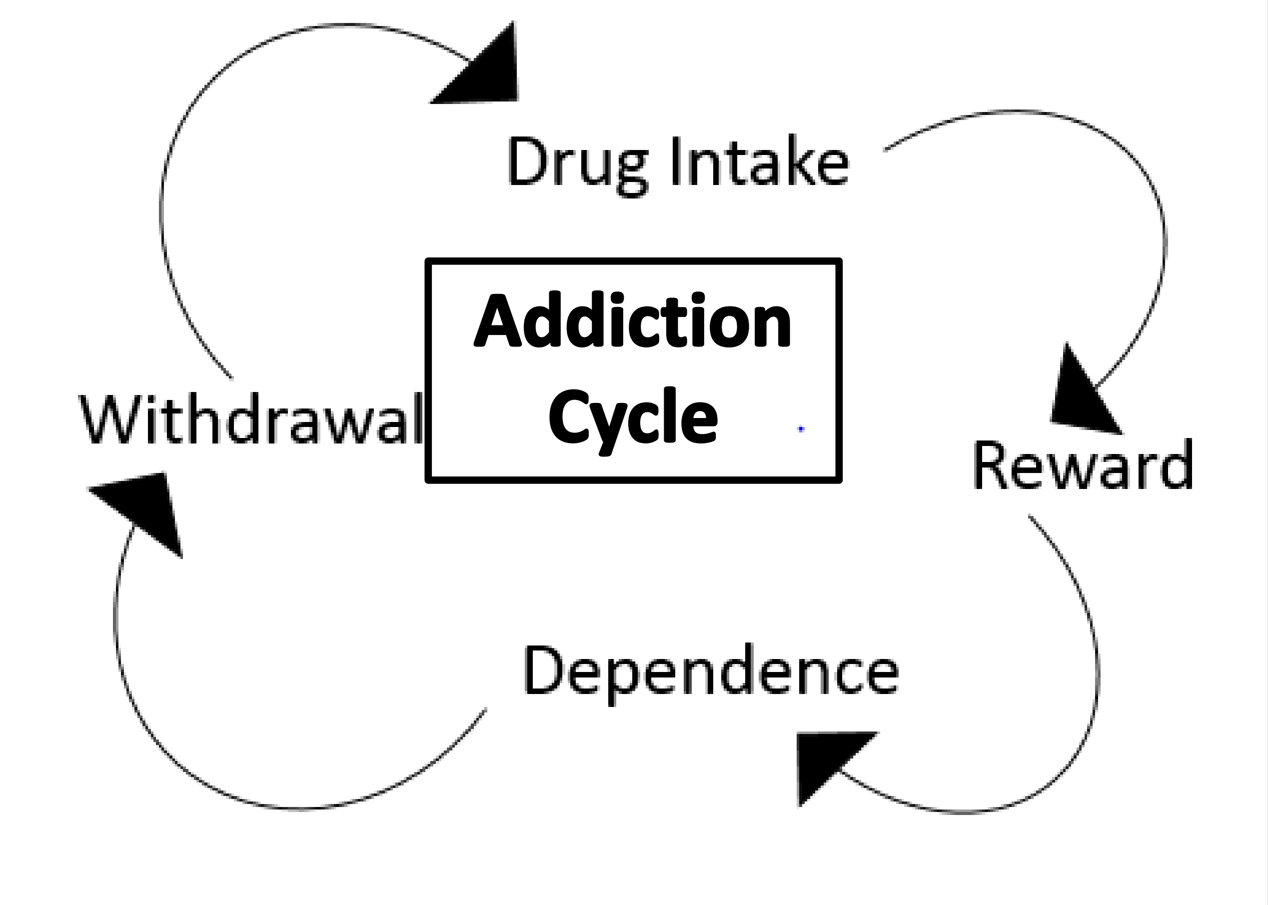
Cost: $740 billion annually related to crime, lost work productivity, and health care (National Institute on Drug Abuse, 2018)



- National Institute of Drug Abuse (NIDA)

- Definitions

- **Addiction**: Addiction is a disease in which, after a period of recreational use, a subset of individuals develops compulsive use that does not stop even in light of major negative consequences. (Nestler and Luscher)
  - **Reward Pathway**
    - Positive reinforcement
  - **Dependence**
    - Psychological
    - Physical
    - Craving
  - **Withdrawal**
    - negative symptoms associated with unfulfilled craving



- Rang & Dale’s Pharmacology 8th edition

- Session Objectives

1. Students will understand the neurobiological basis of the brain's reward pathway
2. Students will become familiar with common classes of drugs of abuse (stimulants, depressants, hallucinogens.)
3. Students will understand how drug-specific pharmacological mechanisms lead to drug abuse
4. Students will be able to identify both general class and drug-specific symptoms of intoxication and withdrawal.

- Reward Pathway: Concepts

- Food
- Water
- Sex
- **Drugs**

Rewarding Stimuli → Reward Pathway

- Activation of Reward Pathway → Increase **Dopamine**
- **“Dopaminergic Reward Pathway”**
  - Dopamine is the messenger that travels from one area of the brain to another, activating pleasure and reward.
  - Repeated drug intake → repeated dopamine stimulation → permanent changes in the brain→ compulsive drug intake

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- Reward Pathway: Anatomy

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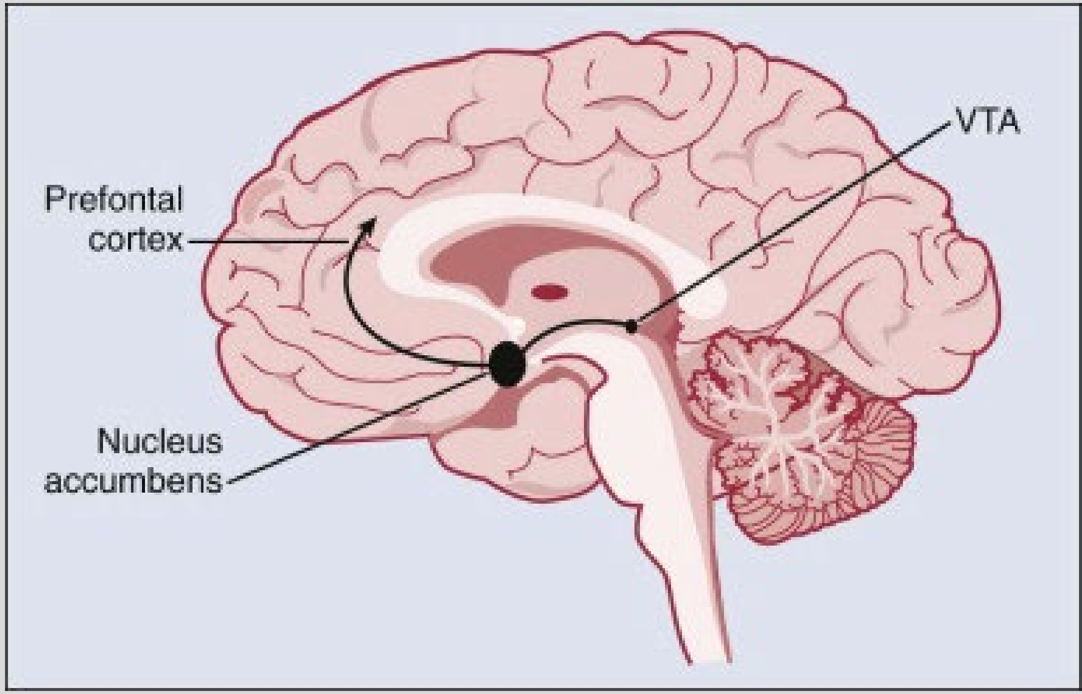
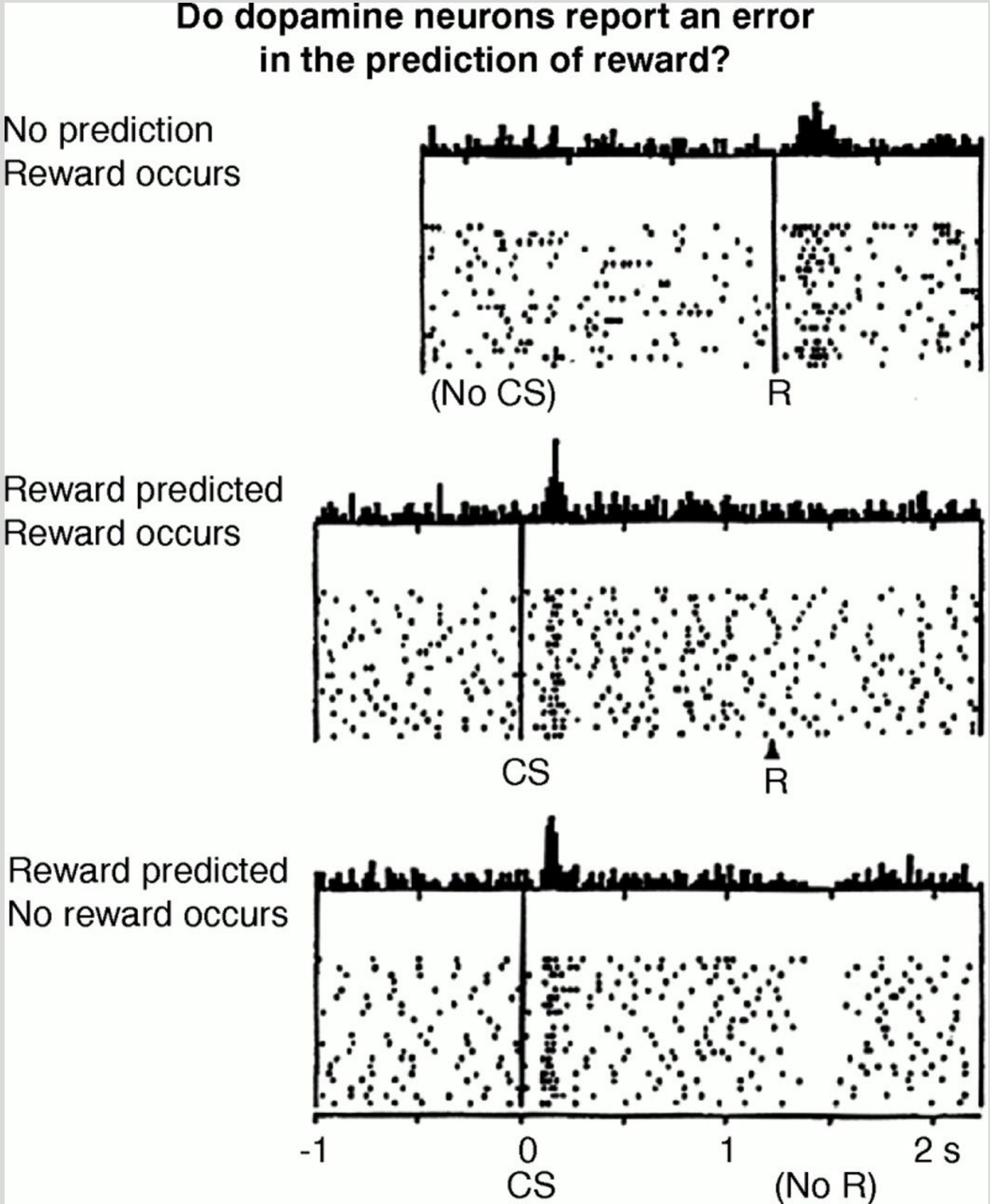
- **“Mesolimbic Dopaminergic Pathway”**
  - Meso in Greek = middle (midbrain)
  - Mesolimbic = Midbrain + Limbic System
- **Midbrain**
  - **Ventral Tegmental Area (VTA): dopamine production**
- **Limbic Area**
  - Amygdala (emotions)
  - Hippocampus (memory)
  - **Nucleus Accumbens** (NA) in basal ganglia (habit formation, pleasure) . It is the place where gene expression alterations produced by excessive dopamine might underlie addiction.
  - Prefrontal Cortex (executive planning)



- Abeloff’s Clinical Oncology, 5th edition

Dopaminergic neurons in VTA encode reward prediction error



Recordings in monkey VTA

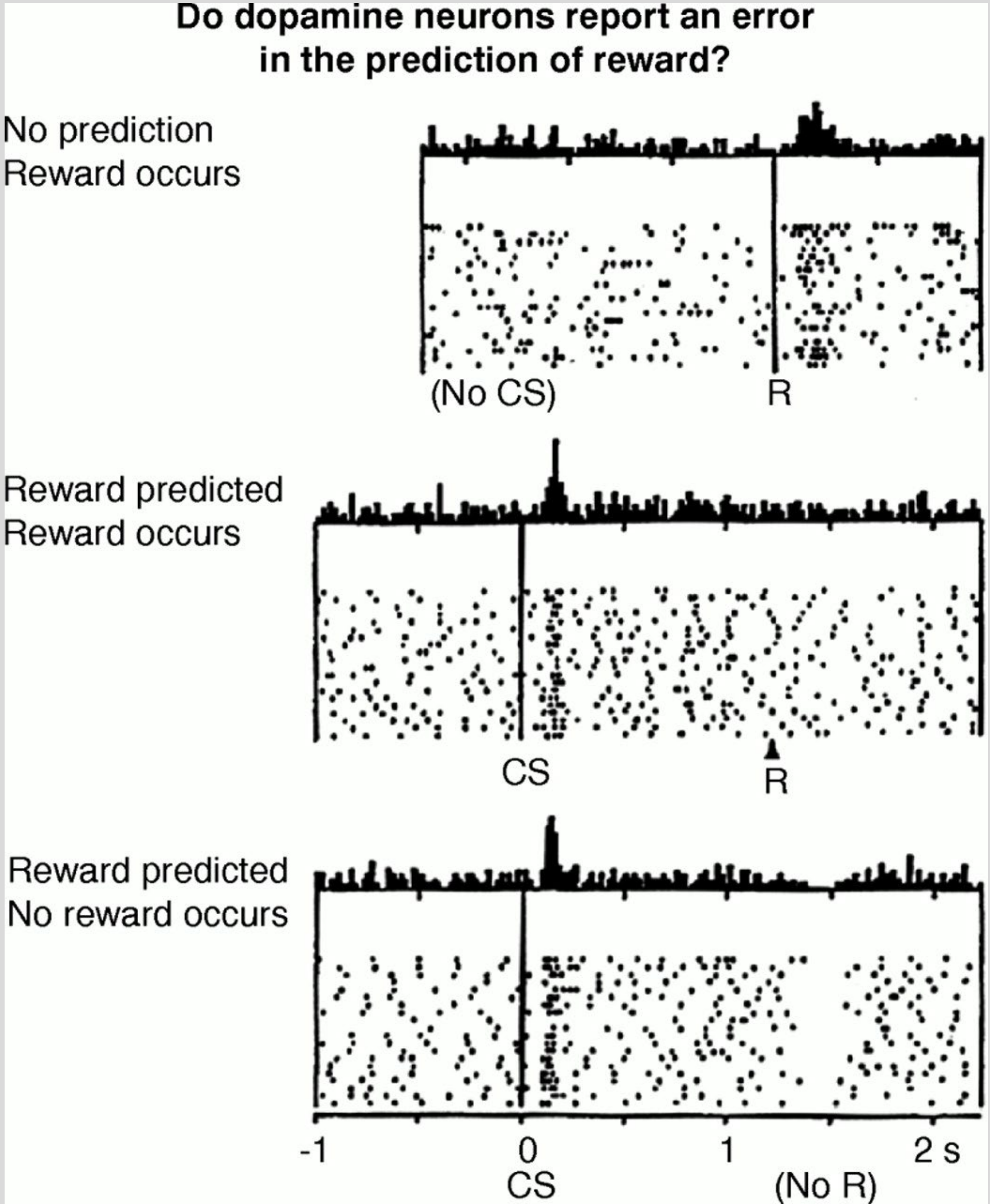
Initially VTA cells increase their firing rate as the monkey receives reward. The reward (R) happens at random times that the monkey cannot predict. Following this unpredictable reward, the VTA was activated.

In the second set of experiments there was a light (Conditioned Stimulis or CS) that preceded the onset of the reward. This light predicts the reward. In this case, VTA neurons responds to the light and it does stop responding to the reward. The idea is that as a stimulus becomes predictable , it stops activating the dopaminergic pathway.

In the third set of experiments the light went on but the reward was omitted. As the reward was omitted the VTA neurons suppressed their spontaneous activity.

A Neural Substrate of Prediction and Reward Wolfram Schultz, Peter Dayan, P. Read Montague*, *Science* 1997

Dopaminergic neurons in VTA encode reward prediction error

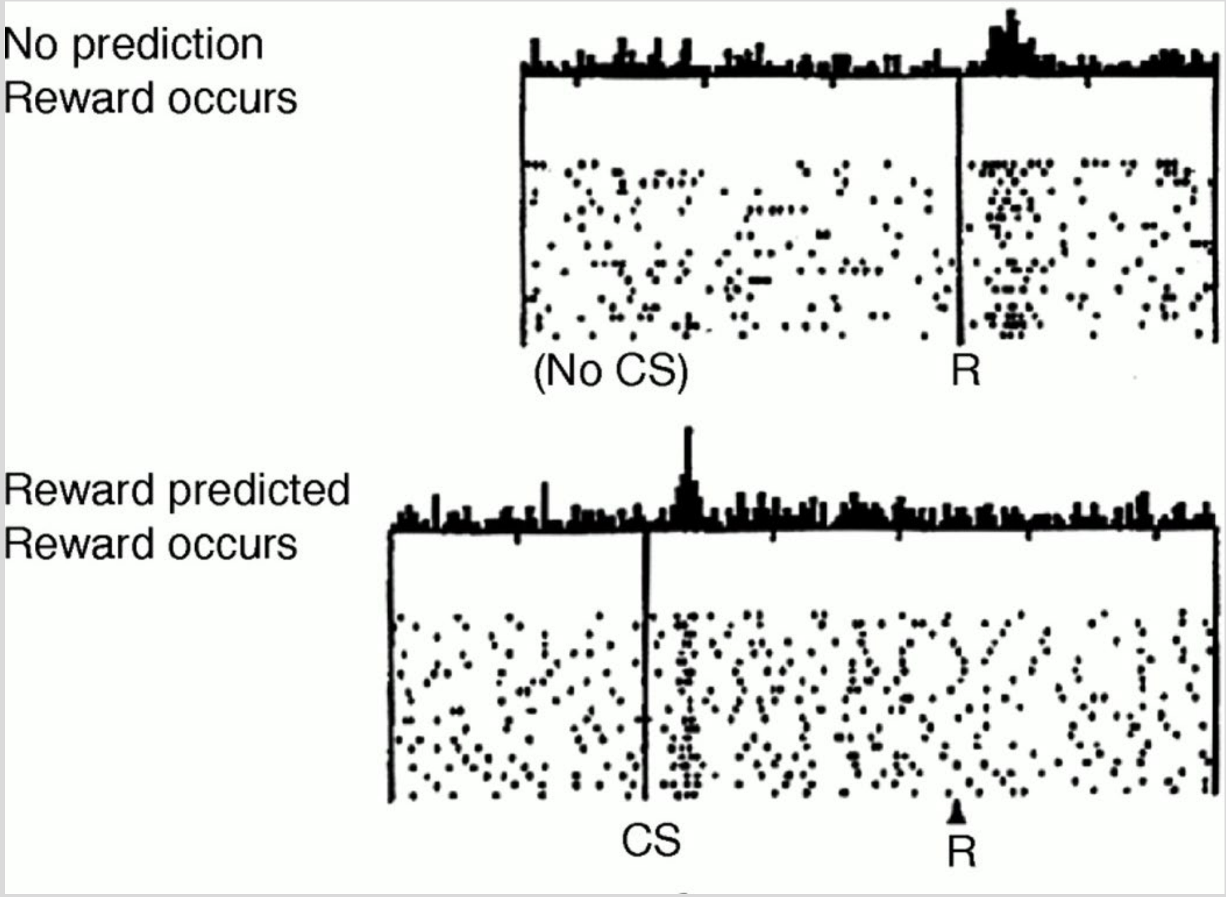


The dopaminergic system in the VTA represents the reward prediction error. If a stimulus is unpredictable, it would produce a large activation in the VTA. However, as a reward becomes more predictable, its activation of VTA is reduced.

Although activation of VTA might be pleasant , the purpose of the system is to learn how to predict high valuable stimuli. This framework of reward prediction error is used in artificial systems that learn to play chess among other applications of a technique called reinforcement learning.

A Neural Substrate of Prediction and Reward Wolfram Schultz, Peter Dayan, P. Read Montague*, *Science* 1997

Drugs of abuse directly affect dopamine transmission



As a natural reward becomes more predictable, it stops the activation of the VTA and the dopaminergic release. However, drugs of abuse directly release dopamine from the axonal terminals of the VTA neurons or directly activate them and do not undergo the natural process of adaptation. They are a much stronger stimuli and there it lies its addiction potential.

A Neural Substrate of Prediction and Reward
Wolfram Schultz, Peter Dayan, P. Read Montague*, *Science* 1997

- **Stimulants**:
  - Speed up CNS activity
  - Mimic affects of endogenous agonists of **sympathetic nervous system**
    - Endogenous agonists = catecholamines (dopamine, norepinephrine, epinephrine)
    - Stimulants act like catecholamines → stimulants are **sympathomimetics**
  - Mechanism of sympathomimetic affect varies, depending on the drug
- **Stimulant Intoxication**: General Symptoms
  - Mood elevation, psychomotor agitation, insomnia, cardiac arrhythmias, tachycardia, anxiety
- **Stimulant Withdrawal**: General Symptoms
  - Post use-**crash**, depression, lethargy, increased appetite, sleep disturbances, vivid nightmares

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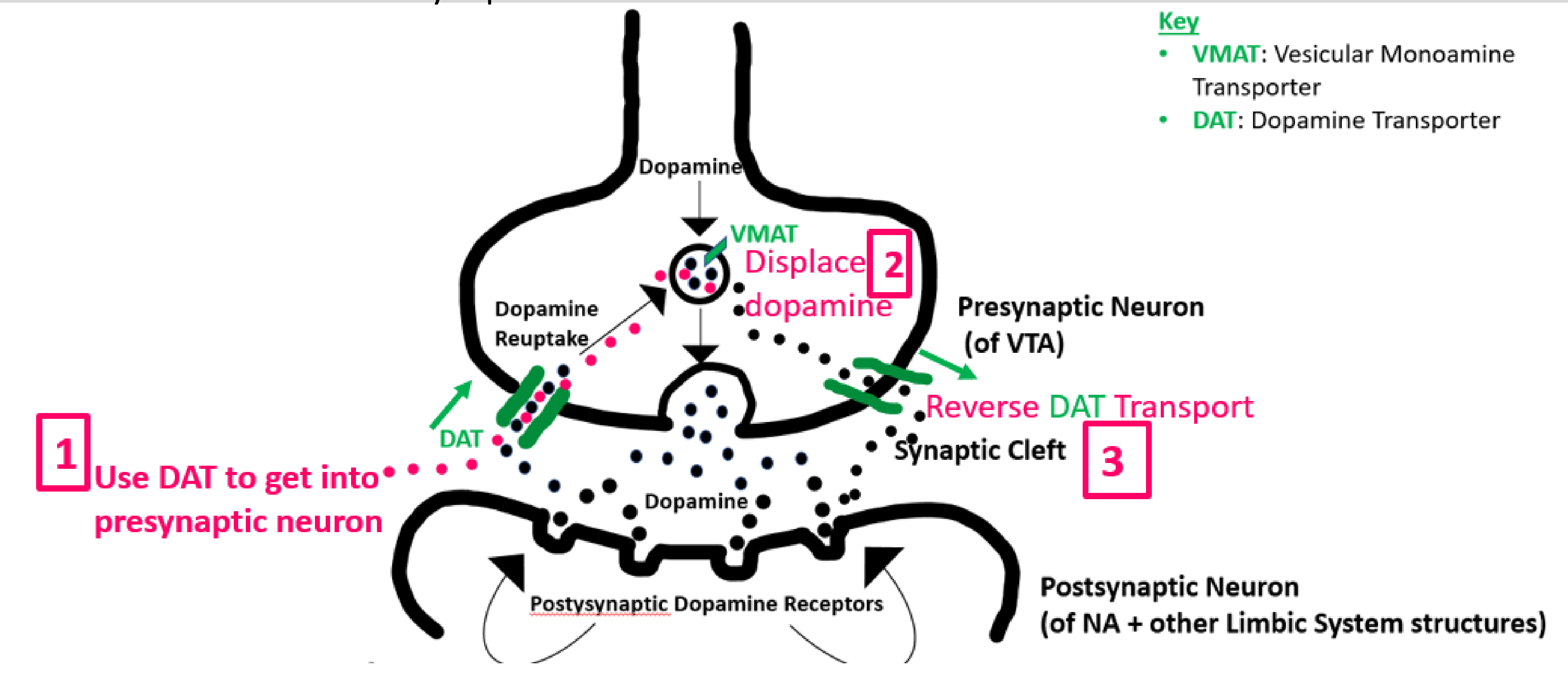
- **Amphetamines**
  - **MOA:**
    - Use **Dopamine Transporter** to get into presynaptic terminal
    - In presynaptic neuron, utilize **VMAT** to enter neurosecretory vesicle, displacing dopamine from vesicle
    - Dopamine concentration in presynatpic neuron reaches threshold, which **reverses action of DAT**
    - Dopamine exits presynaptic neuron via reversed DAT into synaptic cleft
  - **Intoxication:**
    - See General Symptoms
    - Mydriasis (pupillary dilation)
    - Severe: cardiac arrest, seizures
  - **Withdrawal:**
    - See General Symptoms

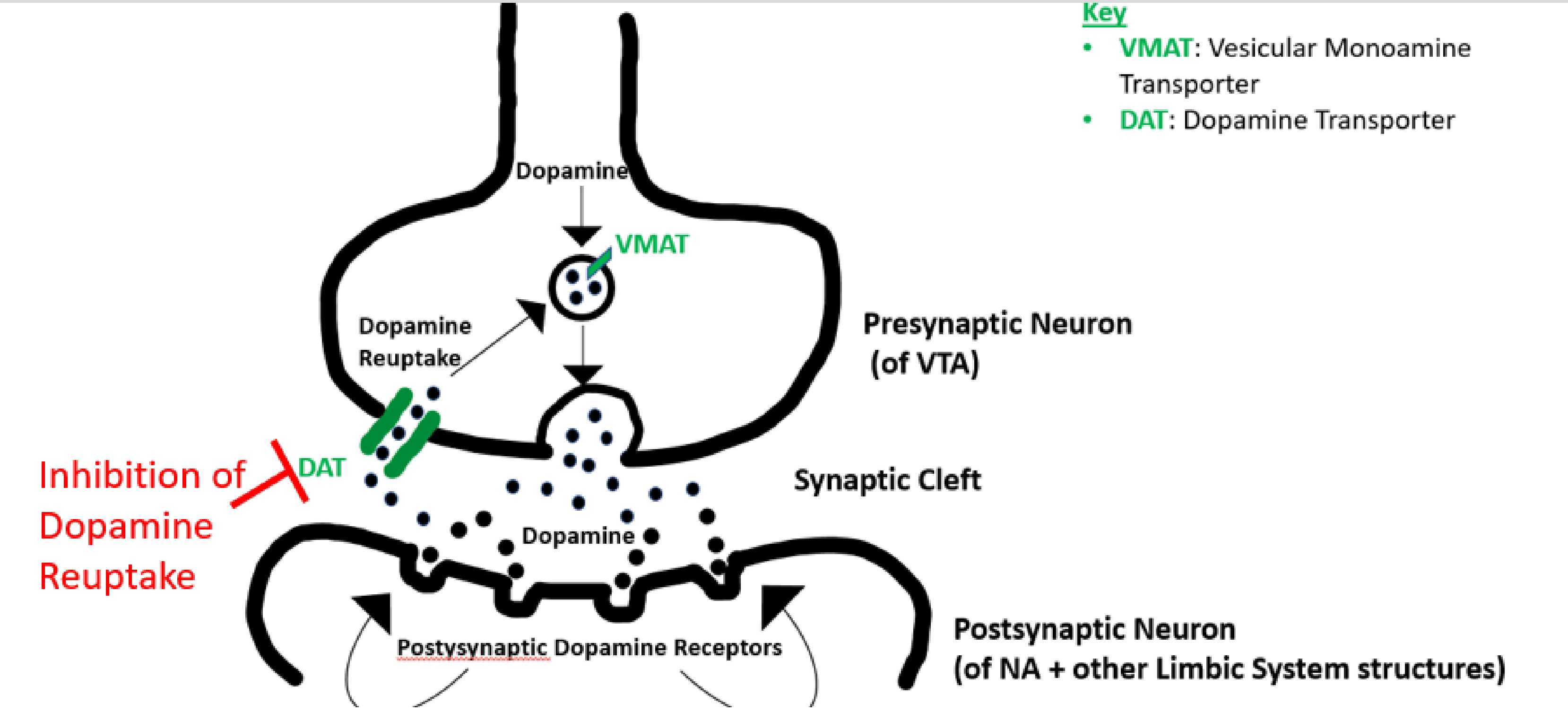


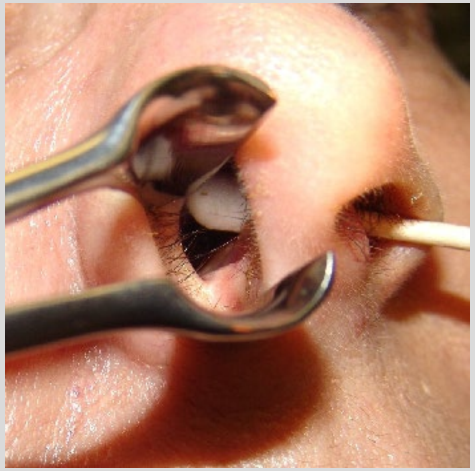
- Rang & Dale's Pharmacology 8th edition

Cocaine

- Inhibits the reuptake of dopamine by inhibiting DAT.
- The inhibition of the dopamine reuptake will produce large activation of the postsynaptic receptors.
- **Coronary artery vasospasm** Mydriasis (pupillary dilation)
- **Hallucinations (tactile)**
- Paranoia
- **Nasal septum perforation produced by tissue death as a consequence of vasospasm.**

- **Withdrawal**:
  - See General Symptoms



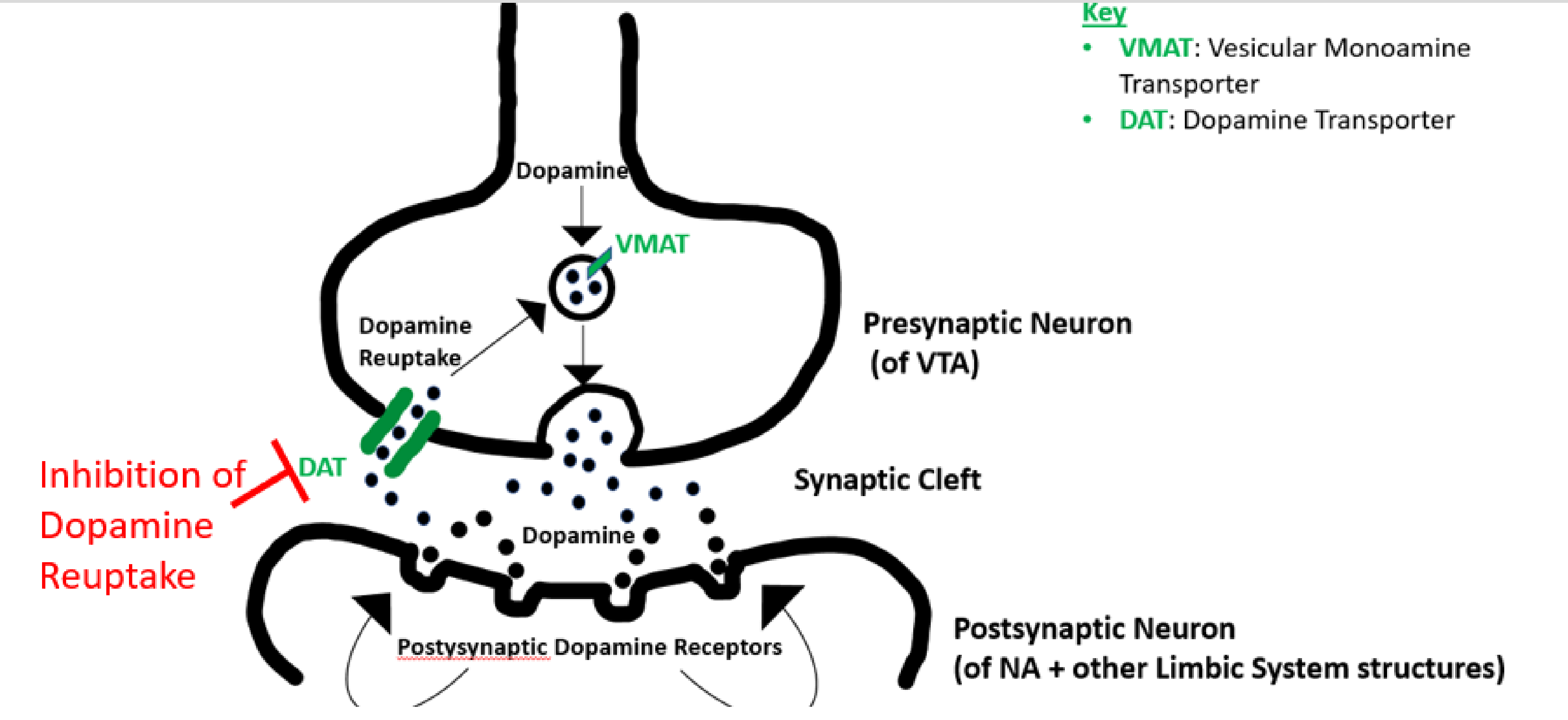


- Goldman-Cecil Medicine, 25th edition
- Meyler's Side Effects of Drugs, 16th edition

- Stimulants: Cocaine

Cocaine

- **MOA**:
  - Inhibits the reuptake of dopamine and **other catecholamines, epinephrine (adrenaline), and norepinephrine**
  - Excess of **catecholamines: Coronary artery vasospasm** →Sudden cardiac death
  - Treatment: alpha blocker or benzo; **do NOT give beta blocker**
  - **If the beta receptors are blocked by the beta blocker, the excess catecholamines will bind to alpha receptors causing further vasoconstriction**

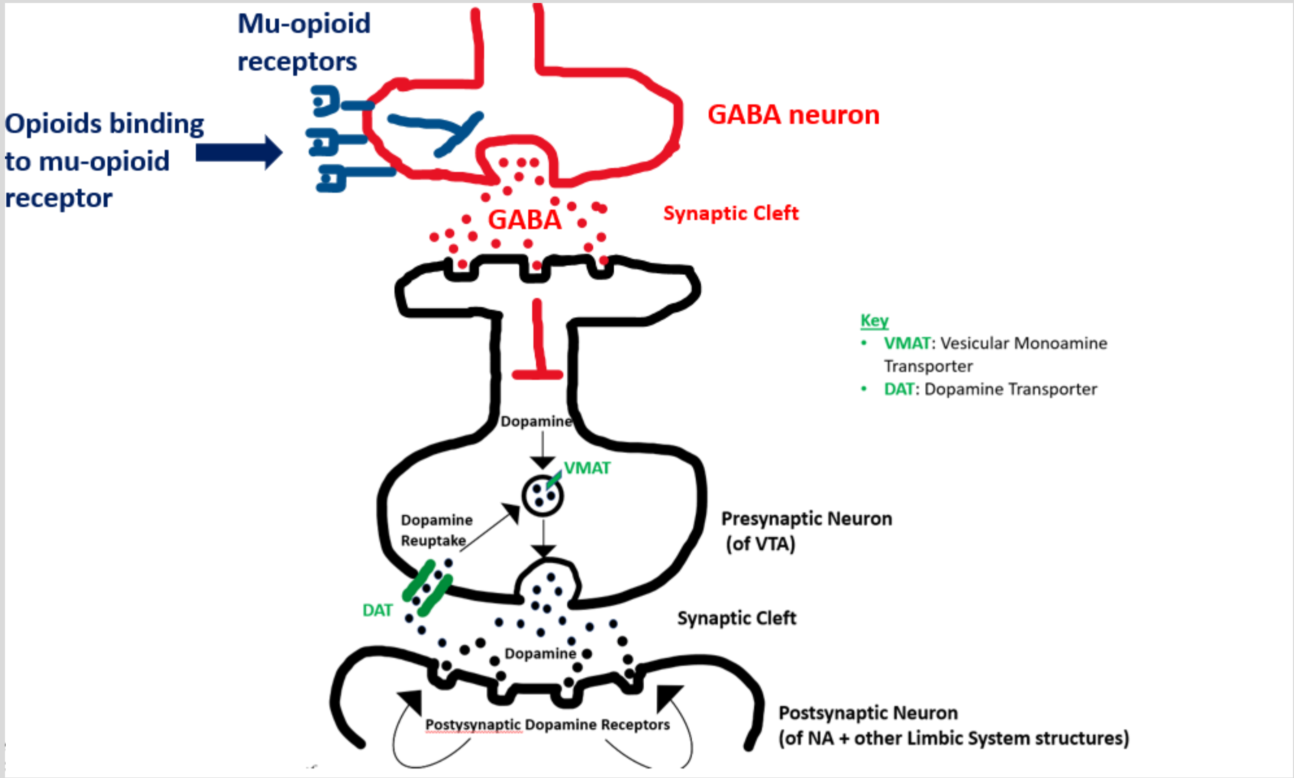


- Goldman-Cecil Medicine, 25th edition
- Meyler's Side Effects of Drugs, 16th edition

• Depressant: Opioids

Opioids

- **MOA:**
  - VTA contains GABA-ergic interneurons that inhibit the dopaminergic neurons.
  - **Opioids bind to mu-opioid receptor in the GABAergic interneurons and inhibit those cells.**
  - There is a reduction of GABA inhibition of the dopaminergic neurons
  - Reduction of inhibitions produces **increased dopamine** release
- **Intoxication**:
  - Respiratory & CNS depression
  - Miosis, **constipation**, itching
- **Withdrawal**:
  - **Diarrhea**, sweating, dilated pupils, piloerection (cold turkey), fever, rhinorrhea, **yawning**, nausea, stomach cramps (flu-like symptoms).
  - Lacrimation is very characteristic of opioid withdrawal.



- An infant born to a mother with a substance use disorder is at risk for withdrawal
- Naloxone is a non-selective and competitive opioid receptor antagonist, used for respiratory depression

- Meyler's Side Effects of Drugs, 16th edition

- Depressant: Alcohol

Alcohol

- **Intoxication**:
  - (**ABCDES**: **a**taxia, **b**lackouts, **c**oma, **d**isinhibition, **e**motional **l**ability, **s**lurred speech)
- **Long Term Abuse Complications:**
  - Alcoholic cirrhosis, hepatitis, pancreatitis, peripheral neuropathy, testicular atrophy
- **Mild Withdrawal:**
  - See general nonspecific
- **Severe Withdrawal:**
    - Autonomic hyperactivity (tachycardia, tremors, anxiety, seizures) & DT
    - **Delirium Tremens**: life-threatening 2-4 day peak after last drink; common in hospital setting (2-4 days post op)
      - Treat DT: benzo- chlordiazepoxide, lorazepam, diazepam
    - **Alcoholic hallucinosis**: distinct condition characterized by **visual hallucinations** 12-48 hrs after last drink

- Emergency Medicine, 2nd edition

- Depressant: Benzodiazepines

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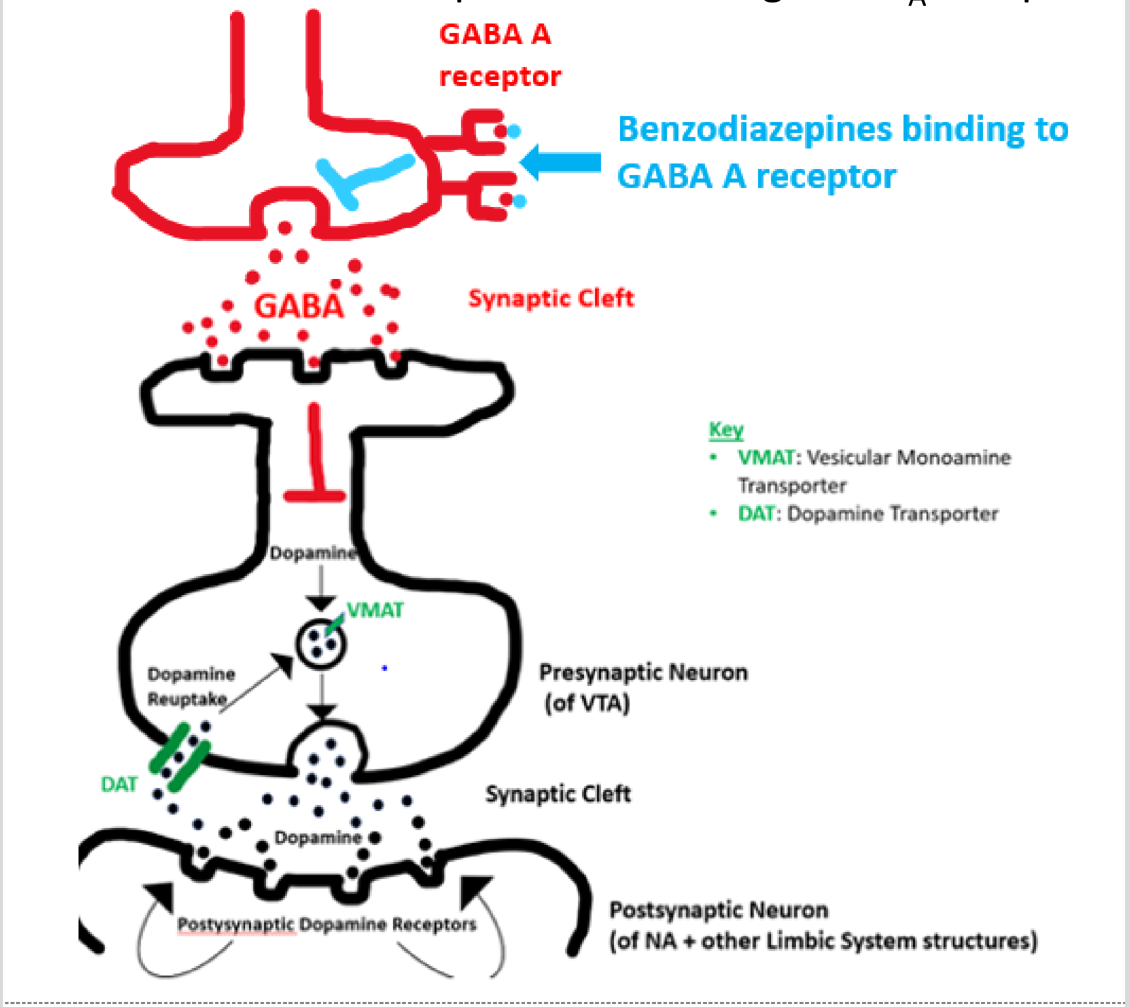
Benzodiazepines

- **MOA:**
  - Binds an allosteric site on GABA A receptor
    - → increases **frequency** of Cl- channel opening → hyperpolarization → inhibitory/depressant affect
      - "*frenzodiazepines* increase *frequency*"
    - → **inhibits GABA interneuron** from releasing GABA → disinhibition dopamine → **increased dopamine** release
- **Intoxication:**
  - Minor resp. depression, **additive CNS affects with alcohol**
    - **Treatment**: flumazenil: competitive antagonist at GABA benzo receptor
- **Withdrawal:**
  - Sleep disturbance, depression, rebound anxiety, seizure

alpha-1-containing $GABA_A$ receptor



- Meyler's Side Effects of Drugs, 16th edition

Barbiturates

- **MOA:**
  - Binds an allosteric site on GABA A receptor
    - → **increases duration** of Cl channel opening
      - "*barbidurates* increase *duration*"
    - → presumably similar reward mechanism as seen in benzos, though not fully understood
- **Intoxication:**
  - Low safety margin**, marked respiratory depression**, **CV depression**, **CNS depression** (exacerbated by alcohol use)
  - **Treatment**: symptom management (assist respiration, increase BP)
- **Withdrawal:**
  - Delirium, life-threatening CV collapse

Meyler's Side Effects of Drugs, 16th edition

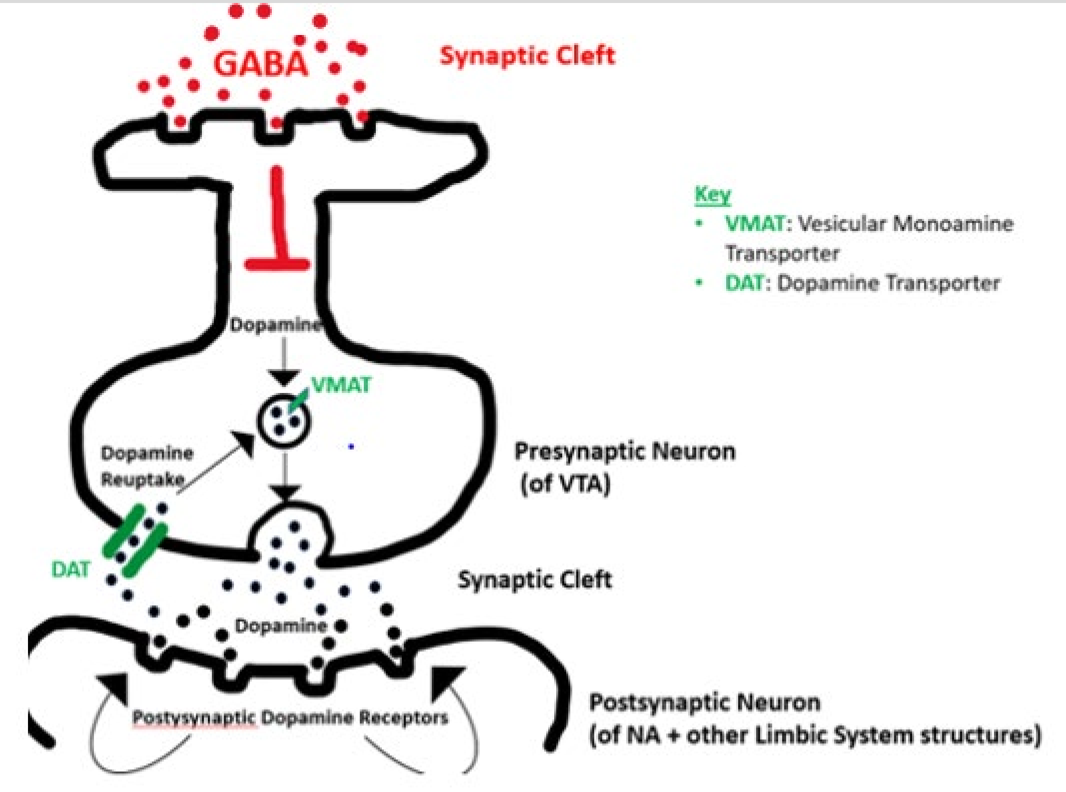
- Hallucinogen: Marijuana (Cannabinoid)

Marijuana

- **MOA:**
  - Active ingredient: **tetrahydrocannibinol (THC)**
  - Binds to and activates **CB1 receptor** on GABA interneuron → inhibits GABA → disinhibits dopamine
- **Intoxication:**
  - Distorted perception of time, **delayed reaction time**, impaired judgement, increased appetite, dry mouth, hallucinations
- **Withdrawal:**
  - Irritability, anxiety, depression, insomnia, restlessness, decreased appetite
  - After discontinuation: detected in **urine 7-10 days** in casual user; **1 month** in chronic user

Marijuana binds to CB1 receptor

endocannabinoid receptor (CB1)



- Goldman-Cecil Medicine, 25th edition

- Conclusions

- The dopaminergic mesolimbic pathway is the neurobiological basis for drug addiction, as it pertains to reinforcing mechanisms
- The three broad classes of drugs of abuse include stimulants, depressants, and hallucinogens, each of which variably target the dopaminergic mesolimbic pathway
- Detailed review of symptomatology, (intoxication/withdrawal clues) guides us in differentiating between drugs of abuse in a clinical setting
- Epidemiological evidence suggests that drug addiction is a serious and growing national problem, making it all the more essential to understand the targeted neuropharmacological pathways